Type MS · Two's complement numbers

Answer 1 of 4 · Question 3 part b.i · 2 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part b.i in this mark scheme extract

3(a)(i)	Award **one** mark for any of the following: • 256 (1) • 2^8 (1) • 2x2x2x2x2x2x2x2 (1)		1
3(a)(ii)	Award **one** mark for each of the following up to a maximum of **three** marks: • 1920 x 1080 in the numerator (1) • x 16/8 (16 in the numerator and 8 in the denominator) or (2 in the numerator) (1) • / (1024 x 1024) in the denominator (1) 1920 pixels x 1080 pixels x 8 bits / 8 bits per byte / 1024 x 1024 bytes per MiB $\frac{1920 \times 1080 \times \frac{16}{8}}{1024 \times 1024}$	Max 2 marks if not a fully correct expression. Allow equivalence e.g. $1024 = 2^{10}$ $1024^2 = 2^{20}$ 2^{-20} is equivalent to $1/2^{20}$	3
3(a)(iii)	**The only correct answer is C** ***A** is not correct because the MSB is 1 so the denary value must be more than 128* ***B** is not correct because the LSB is 0 so the denary value must be even* ***D** is not correct because the denary number is signed*		1

Type MS · Two's complement numbers

Answer 1 of 4 · Question 3 part b.i · 2 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part b.i in this mark scheme extract

3(b)(i)	Award **one** mark for each of: • Negative sign (1) • 63 (1) Example: - 63		2
3(b)(ii)	Award **one** mark for: 1000 0010 (1)		1
3(b)(iii)	Award up to **two** marks for a linked explanation such as: • An overflow error has occurred (1) because the result requires more bits to store than are available (1)		2
3(c)(i)	Award **one** mark for any of the following: • 011 0101 (1) • 53 (1) • Hex value of 35 (1) Do not accept 11 0101 (six bits, ASCII is at least seven)		1

Type MS · Two's complement numbers

Answer 1 of 4 · Question 3 part b.i · 2 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part b.i in this mark scheme extract

3(c)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • It is a standard/universal encoding method (1) • It can represent all the characters in the English alphabet (1) • It only uses 7/8 bits **per character** (1) • It can represent 128/256 characters (in standard/extended format) (1) • It takes up less storage space than Unicode (1) • It can be used with bit masking (1) • It is compatible with UTF-8 (1)		2
		Total for question 3	**13**

Type MS · Two's complement numbers

Answer 1 of 4 · Question 3 part b.i · 2 marks

2024 JUN · Paper 01 · extract 4 of 4 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Two's complement numbers

Answer 2 of 4 · Question 2 part a.ii · 2 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part a.ii in this mark scheme extract

2(a)(i)	1101 1110 Award one mark for: • MSB = 1 (1) • Rest of pattern correct (1)		2

Question Number	Answer	Additional Guidance	Mark
2(a)(ii)	- (1) 119 (1)		2

Question Number	Answer	Additional Guidance	Mark
2(b)	B 16		1

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	128	Allow the range 0-127	1

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	C		1

Question Number	Answer	Additional Guidance	Mark

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

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Answer 2 of 4 · Question 2 part a.ii · 2 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part a.ii in this mark scheme extract

2(c)(iii)	Award **two** marks for a linked explanation such as: (Before Unicode existed) there are hundreds of different encoding systems (1) and no single encoding system could contain enough characters to represent all major languages (1) Standard ASCII only provides 128 different patterns (1) can't represent all major languages/symbols/characters (1) Unicode uses a minimum of 16 bits (1) so can represent at least 2^{16} characters (1) Unicode has very large number of characters (1) can represent all languages/ASCII was developed (just) for English (1)		2

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · Two's complement numbers

Answer 2 of 4 · Question 2 part a.ii · 2 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Two's complement numbers

Answer 3 of 4 · Question 1 part a.ii · 2 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part a.ii in this mark scheme extract

1(ai)	0100 1101 100 1101 Award **one** mark for each correct nibble	Must be in the correct order	2
1(a)(ii)	1000 0010 Award **one** mark for each correct nibble	Must be in the correct order	2
1(a)(iii)	Award **one** mark for each of: • C (1) • 6 (1)	Must be in the correct order	2

Type MS · Two's complement numbers

Answer 3 of 4 · Question 1 part a.ii · 2 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	400 pixels x 200 pixels x 12 bits / 8 bits per byte Allow expression in word form $\frac{400 \times 200 \times 12}{8}$ 400 x 200 **OR** 80000(1) 12/8 **OR** 1.5 (1)	Accept any other use of 400, 200, 12, 8 that gives the correct answer Allow one mark for 120,000 with no expression	2
1(b)(ii)	Spaces are to help legibility. Award **three** marks for: Line 1 3W 1B 4W Line 2 1W 6B 1W Line 3 1B 2W 1B 1W 1B 2W Line 4 3W 2B 3W Award **two** marks for any three lines correctly encoded Award **one** mark for any two lines correctly encoded	Allow **one** error in the sequence Accept letter before number but must be consistently used	3
1(b)(iii)	Award **one** mark from: • None of the original (image) detail/quality is lost (when the image is stored) (1) • None of the data/information is (permanently) removed (1) • It can be decompressed without losing detail (1)		1

Type MS · Two's complement numbers

Answer 3 of 4 · Question 1 part a.ii · 2 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	Award up to **three** marks for a linked description: • set the sample rate/parameters/bit-depth (1) • sample (the analogue sound) (1) • measure the sound amplitude/volume/frequency (1) • give a (binary) value/number for each measurement (1) • store data as sample rate and values / digital signals (1)		3
1(c)(ii)	Award **one** mark from: • sound may be missing some frequencies (1) • some audio information/data is lost/deleted during the (compression) process (1)		1
		Total	**16**

Type MS · Two's complement numbers

Answer 3 of 4 · Question 1 part a.ii · 2 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Two's complement numbers

Answer 4 of 4 · Question 2 part d.iii · 2 marks

2023 JUN · Paper 01 · extract 1 of 5 · find part d.iii in this mark scheme extract

<table>
<tr>
<td>2(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Denary</th><th>Hexadecimal</th></tr>
<tr><td>8</td><td>8 (1)</td></tr>
<tr><td>12</td><td>C (1)</td></tr>
</table>
</td>
<td></td>
<td>2</td>
</tr>
</table>

Type MS · Two's complement numbers

Answer 4 of 4 · Question 2 part d.iii · 2 marks

2023 JUN · Paper 01 · extract 2 of 5 · find part d.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(b)	**The only correct answer is D** ***A*** *is not correct because* 6^2 *stores 36 patterns, not 64 patterns* ***B*** *is not correct because* **6 × 2** stores 12 patterns, not 64 patterns ***C*** *is not correct because* 6^2 **– 1** stores 35 patterns, not 64 patterns		1

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Award **one** mark for: • 0001 0111 (1)	Ignore spaces	1

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	Award **one** mark for: • 1111 1000 (1)	Ignore spaces	1

Type MS · Two's complement numbers

Answer 4 of 4 · Question 2 part d.iii · 2 marks

2023 JUN · Paper 01 · extract 3 of 5 · find part d.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	Award **one** mark for each correct nibble: • 0011 1100	Ignore spaces	2

Question Number	Answer	Additional Guidance	Mark
2(d)(ii)	Award **one** mark for each of: • Negative sign (1) • 22 (1) Example: • -22	Note 22 is calculated from 16 + 4 + 2	2

Question Number	Answer	Additional Guidance	Mark
2(d)(iii)	Award **one** mark for each correct nibble: • 0001 1011	Ignore spaces	2

Type MS · Two's complement numbers

Answer 4 of 4 · Question 2 part d.iii · 2 marks

2023 JUN · Paper 01 · extract 4 of 5 · find part d.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(e)	Award **one** mark for each of: 1. 13 x 1000 // 13000 in the numerator (1) 2. 1024 in the denominator (1) Examples: $\frac{13 \times 1000}{1024}$ $\frac{13000}{1024}$ $\frac{13 \times 500}{512}$ $\frac{13 \times 250}{256}$ $\frac{13 \times 125}{128}$ $\frac{13 \times 10^3}{2^{10}}$	Award reduced expressions, if correct Allow all equivalent exponents Note 1000/1024 = 1.024, so could have 13 * 1.024	2

Type MS · Two's complement numbers

Answer 4 of 4 · Question 2 part d.iii · 2 marks

2023 JUN · Paper 01 · extract 5 of 5 · find part d.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark